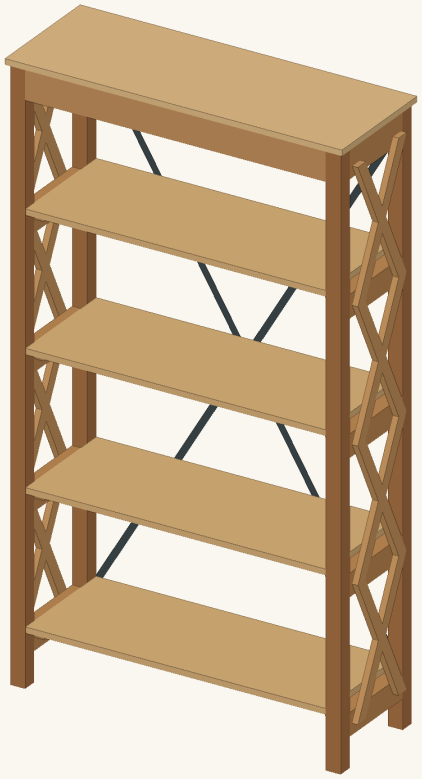
Alderline X4

FOUR-SHELF X-FRAME BOOKSHELF • ORIGINAL WORKSHOP PLAN

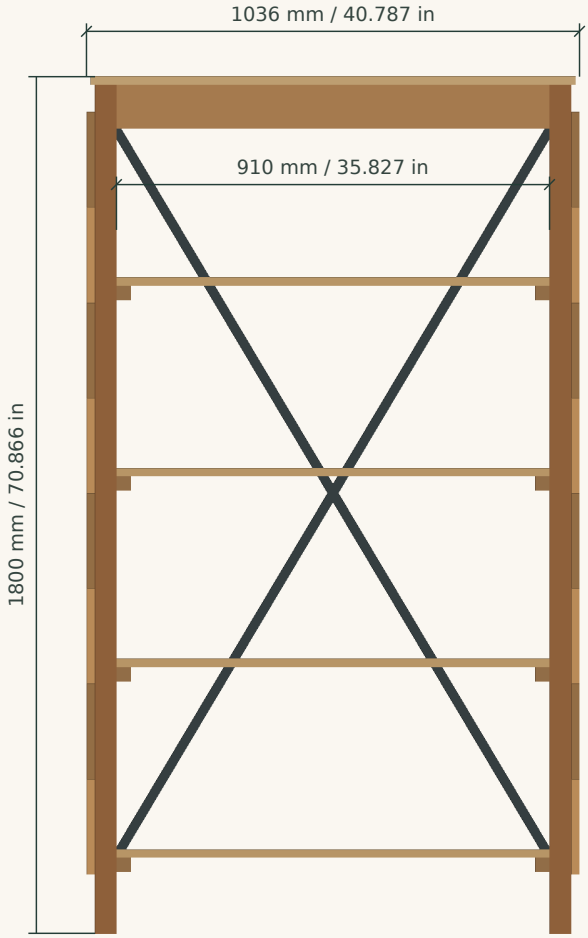
An open bookshelf with timber posts, repeated half-lapped side Xs, plywood shelves and a separate top. Rear steel cross-straps brace the open back. This is a prototype plan: build and test before commercial release.



Maximum size: 1036 mm / 40.787 in wide × 370 mm / 14.567 in deep × 1800 mm / 70.866 in high. Intermediate skill; dowel jig, controlled half-lap routing and two-person final assembly.

Metric dimensions control; imperial values are rounded decimal-inch equivalents. No safe working load has been established. Wall restraints are required before loading.

Measured front elevation



Frame width: 1000 mm / 39.370 in

Opening: 910 mm / 35.827 in

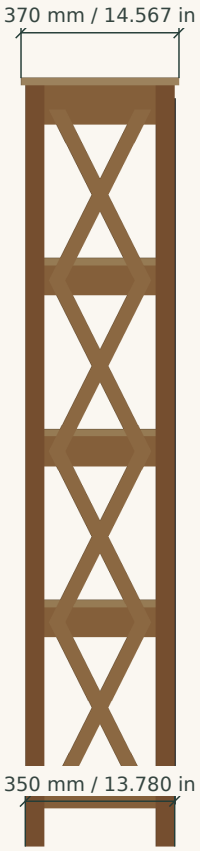
Shelf length: 909 mm / 35.787 in

Top width: 1020 mm / 40.157 in

Overall width: 1036 mm / 40.787 in

The X slats project 18 mm / 0.709 in beyond each frame side. Height excludes foot pads and wall restraint fittings.

Side geometry & shelf levels



Shelf	Underside above floor	Top above floor
1	160 mm / 6.299 in	178 mm / 7.008 in
2	560 mm / 22.047 in	578 mm / 22.756 in
3	960 mm / 37.795 in	978 mm / 38.504 in
4	1360 mm / 53.543 in	1378 mm / 54.252 in

Shelf bearing on each cleat: 29.5 mm / 1.161 in wide × 260 mm / 10.236 in long. Side clearance: 0.5 mm / 0.020 in each; front/rear setback: 1 mm / 0.039 in each.

Cut list / finished parts

Thickness × width × length. Use actual finished stock. Inspect plywood bearing edges for voids; orient face grain along the shelf length. H length is along either long edge, not the rough blank.

ID / QTY	Part & material	Finished dimensions	Cutting detail
A / 4	Posts Hardwood	45 × 45 × 1782 mm 1.772 × 1.772 × 70.157 in	Square ends; matched finished stock.
B / 8	Shelf side rails Hardwood	45 × 70 × 260 mm 1.772 × 2.756 × 10.236 in	Two rail dowels per end.
C / 2	Upper side rails Hardwood	45 × 92 × 260 mm 1.772 × 3.622 × 10.236 in	Two rail dowels per end.
D / 4	Shelf panels Cabinet plywood	18 × 348 × 909 mm 0.709 × 13.701 × 35.787 in	Face grain along span; no notches.
E / 1	Top panel Cabinet plywood	18 × 370 × 1020 mm 0.709 × 14.567 × 40.157 in	Overhang from frame: 10 mm / 0.394 in.
F / 2	Front/rear top aprons Cabinet plywood	18 × 92 × 910 mm 0.709 × 3.622 × 35.827 in	Two apron dowels per end.
G / 8	Shelf support cleats Hardwood	30 × 30 × 260 mm 1.181 × 1.181 × 10.236 in	Transverse screws into side rails.
H / 16	X slats Hardwood	18 × 35 × 447.59 mm 0.709 × 1.378 × 17.622 in	26.66 deg off square; 9 mm / 0.354 in lap; length along long edge.
J / 2	Rear steel straps Bracing steel	1 × 20 × 1913.34 mm 0.039 × 0.787 × 75.328 in	End-hole pitch 1863.34 mm / 73.360 in; measure on squared case.

Hardware & joint schedule

Installed quantities below are counted from connection sites. Buy spare fasteners and test actual pilot sizes in matching scrap. Purchased products remain to be selected and verified.

ID / QTY	Hardware	Allocation
K1 / 40	Fluted dowel 8 × 50 mm	Two per rail end into A.
K2 / 8	Fluted dowel 6 × 40 mm	Two per apron end into A.
K3 / 24	wood screw 5 × 60 mm	3 per G into B.
K4 / 16	wood screw 4 × 40 mm	4 per D into G.
K5 / 32	wood screw 4 × 45 mm	One at each H end into B/C.
K6 / 4	Pan-head wood screw 5 × 35 mm	One per J end into rear posts.
K7 / 2	Washer 1 mm thick; clearance for K6	Behind both ends of J-2.
K8 / 4	Steel angle 25 × 25 × 20 mm; one hole/wing	Two on each F, supporting E.
K9 / 8	wood screw 4 × 16 mm	One per angle wing.
K10 / 2	Commercial furniture-to-wall restraint kit	One at each rear post; product-specific wall fixings.
K11 / 4	Equal non-slip foot pad	One per A foot.

Use wood adhesive appropriate to the selected timber and follow its cure instructions. Top angles assume two steel wings with one hole each. Do not substitute wall fixings without checking the actual wall and chosen restraint system.

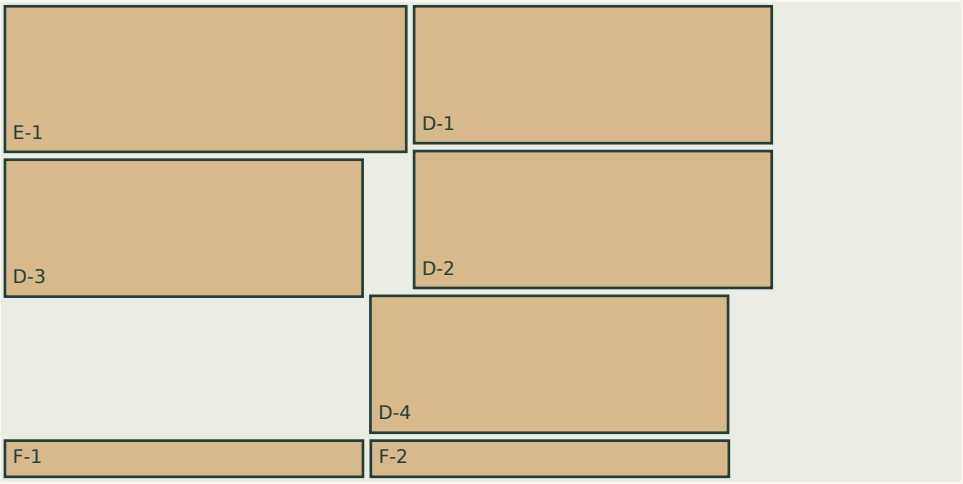
Material purchasing & plywood

Layouts are feasible purchasing maps, not optimized saw sequences. They reserve end trims and kerf; buy additional stock for knots, tear-out and defects.

Stock	Quantity	Purchase basis
A	4	Hardwood 45 x 45 x 2400 mm; add defect allowance
B-C	2	Hardwood 45 x 92 x 2400 mm; add defect allowance
G	1	Hardwood 30 x 30 x 2400 mm; add defect allowance
H	4	Hardwood 18 x 35 x 2400 mm; add defect allowance
PLY	1	Cabinet plywood 18 x 1220 x 2440 mm
STEEL	2	Bracing strip 1 x 20; rough length at least 1933.3 mm

Crosscut kerf 3 mm / 0.118 in; end trim 10 mm / 0.394 in; sheet separation at least 20 mm / 0.787 in. All plywood grain runs horizontally in the maps.

PLY-1 / 2440 mm / 96.063 in × 1220 mm / 48.031 in / grain →



Geometric waste incl. kerf/trim/offcuts: 39.2%. Map is not a saw sequence.

Lumber cutting maps

Each band represents one purchased board. Dimensions label rough blank lengths. B and C share wider stock; rip B to its listed height after crosscutting. H blanks include angled-end allowance.

A-1 / stock 45 × 45 × 2400 mm



A-2 / stock 45 × 45 × 2400 mm



A-3 / stock 45 × 45 × 2400 mm



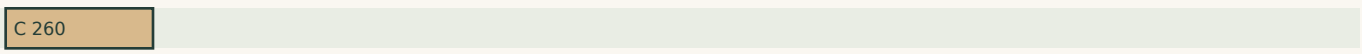
A-4 / stock 45 × 45 × 2400 mm



B-C-1 / stock 45 × 92 × 2400 mm



B-C-2 / stock 45 × 92 × 2400 mm



G-1 / stock 30 × 30 × 2400 mm



H-1 / stock 18 × 35 × 2400 mm



H-2 / stock 18 × 35 × 2400 mm



H-3 / stock 18 × 35 × 2400 mm



H-4 / stock 18 × 35 × 2400 mm

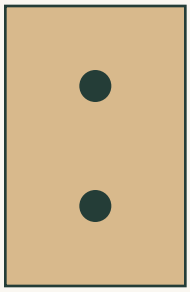


Residual material includes crosscut kerfs, end trims and offcuts. Ripping waste and defects are not quantified. Never use a small offcut where safe clamping or a guarded cut cannot be maintained.

Dowel joints / enlarged end views

B END / Ø8 mm

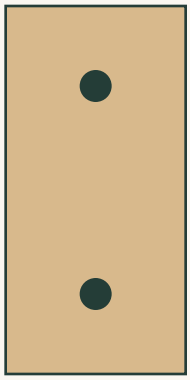
45 × 70 mm



Bores centred in thickness

C END / Ø8 mm

45 × 92 mm



Bores centred in thickness

F END / Ø6 mm

18 × 92 mm



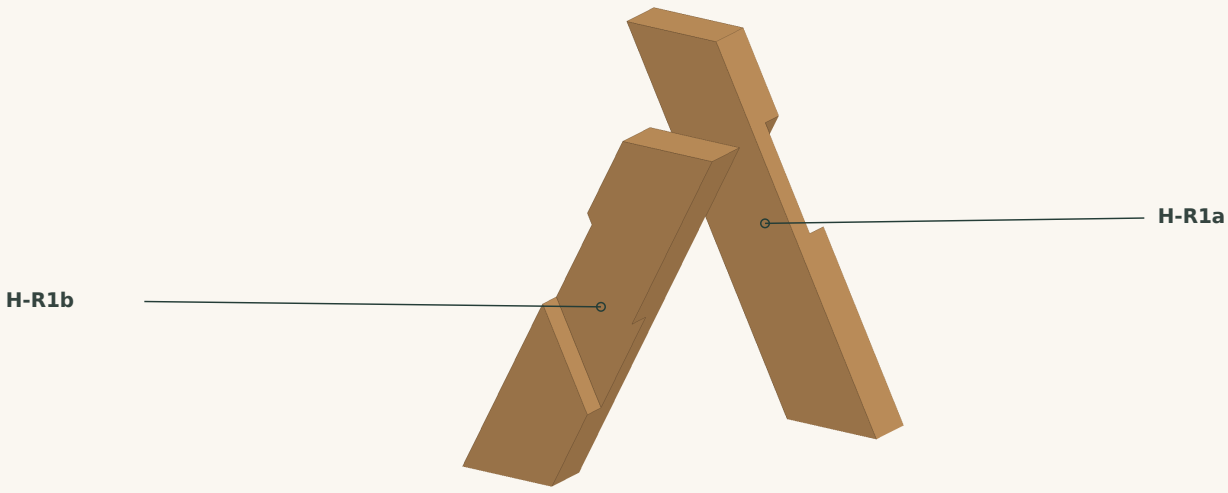
Bores centred in thickness

Joint	Member bottom level	Post bore heights from foot
B station 1	90 mm / 3.543 in	110 / 140 mm
B station 2	490 mm / 19.291 in	510 / 540 mm
B station 3	890 mm / 35.039 in	910 / 940 mm
B station 4	1290 mm / 50.787 in	1310 / 1340 mm
C and F	1690 mm / 66.535 in	1710 / 1762 mm

Side rail/post mating bores: Ø8, nominal embed 25 mm / 0.984 in per member. Apron/post bores: Ø6, nominal embed 20 mm / 0.787 in per member. Drill each blind hole 1 mm / 0.039 in deeper than nominal embed using a stop.

A side-rail bores are on the front/rear post faces facing the side rails, centred 22.5 mm / 0.886 in across the face. Apron bores face across the opening, 9 mm / 0.354 in from the front/rear outside post edge. Mirror left/right and front/rear posts. Use one finished foot datum for every height. Test the complete joint on scrap before drilling parts.

The half-lapped X / close-up

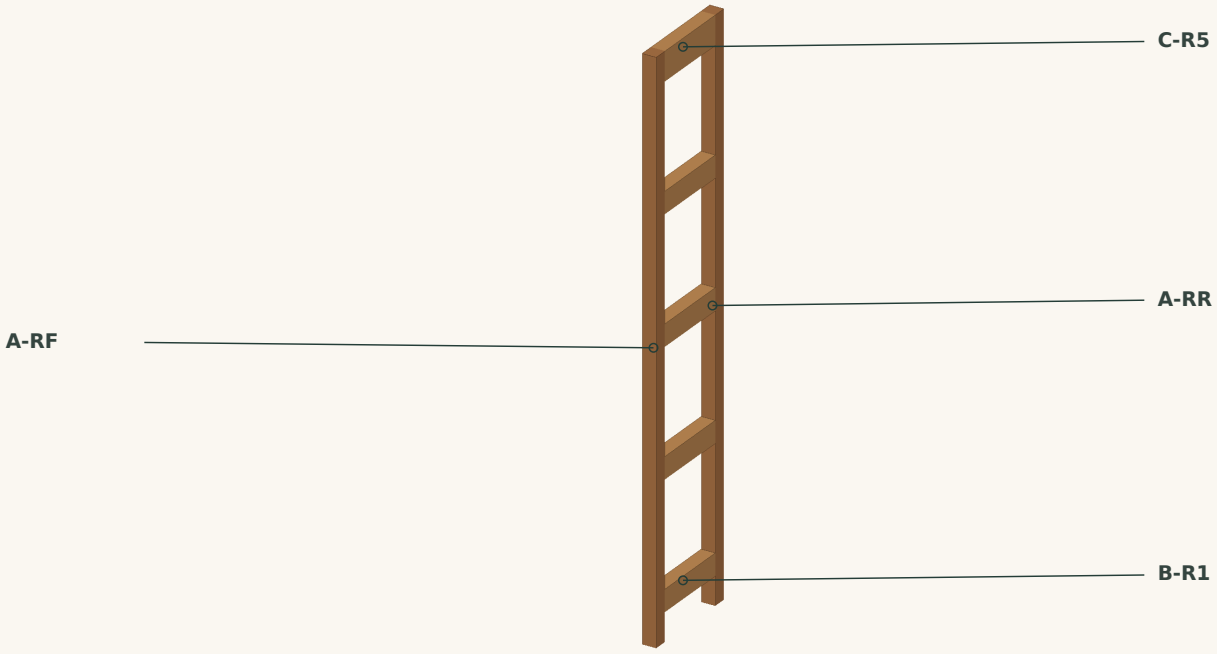


Complementary laps remove 9 mm / 0.354 in from opposite faces. The overlap is a rhombus with 43.64 mm / 1.718 in along either long edge; its boundaries parallel the mating slat edges. The close-up separates the members for inspection.

End cut: 26.66° off square. Finished long-edge length 447.59 mm / 17.622 in; minimum projected extremity length 465.16 mm / 18.313 in. Start from 490 mm / 19.291 in rough blanks. The horizontal end-cut projection is 39.16 mm / 1.542 in.

Use a full-size scrap bay jig. Mark the true crossed overlap, route/saw waste in shallow controlled passes, and keep faces flat. Test for a flush, non-forced fit before repeating. The supplied 1:1 A4/Letter PDFs must be printed at actual size and calibrated with a ruler; they are patterns, not safe router fixtures.

01 / Build the two matched sides

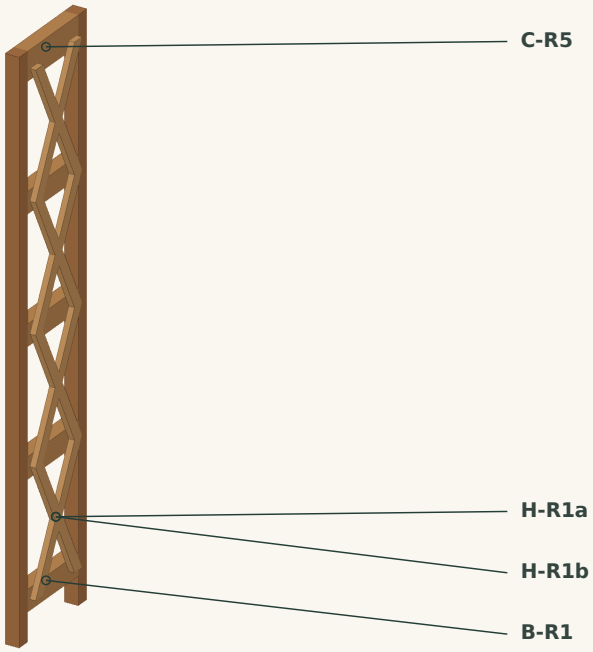


Part identifiers: A-LF, A-LR, B-L1, B-L2, B-L3, B-L4, C-L5, A-RF, A-RR, B-R1, B-R2, B-R3, B-R4, C-R5

Hardware for this step: K1 ×40

Label left/right and front/rear posts viewed from the front. Set side rail bottoms at 90 / 490 / 890 / 1290 mm from the finished feet; upper rail starts at 1690 mm / 66.535 in. Seat two 8 mm / 0.315 in dowels per rail end. Rail end bores are thickness-centred; lower rail height offsets are 20 / 50 mm; upper offsets 20 / 72 mm. Drill 26 mm / 1.024 in deep for 25 mm / 0.984 in nominal embed in each member. Dry-fit, glue, clamp flat and square; matched side depth 350 mm / 13.780 in.

02 / Fit the repeated exterior Xs

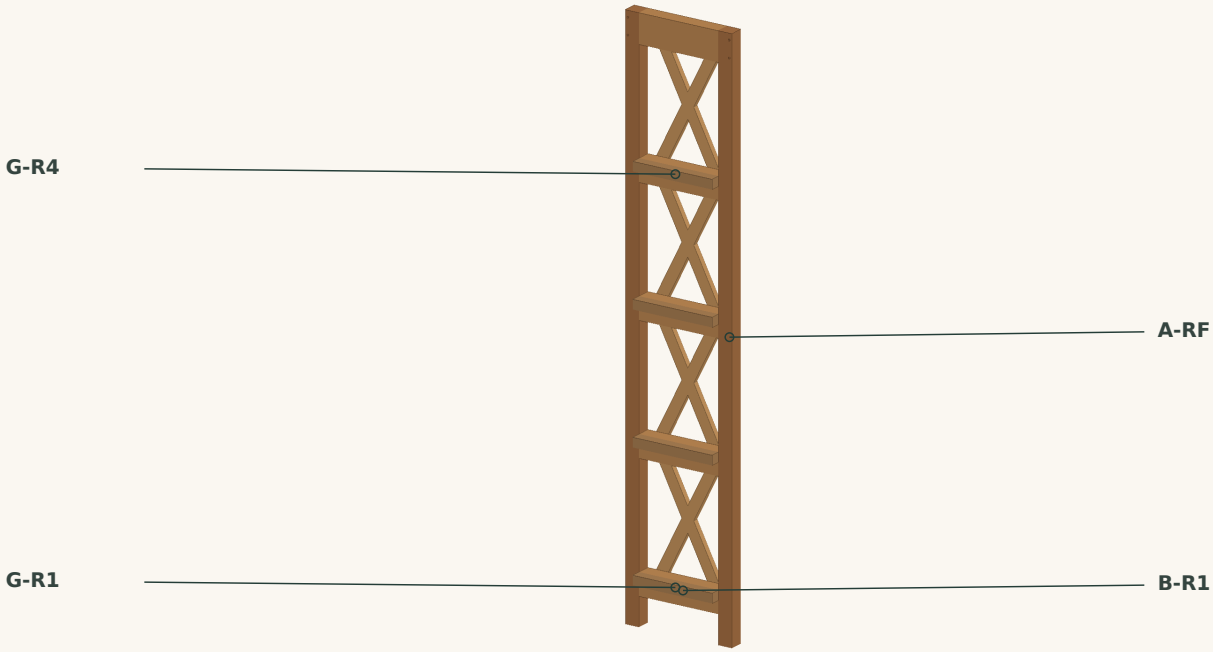


Part identifiers: A-LF, A-LR, B-L1, B-L2, B-L3, B-L4, C-L5, H-L1a, H-L1b, H-L2a, H-L2b, H-L3a, H-L3b, H-L4a, H-L4b, A-RF, A-RR, B-R1, B-R2, B-R3, B-R4, C-R5, H-R1a, H-R1b, H-R2a, H-R2b, H-R3a, H-R3b, H-R4a, H-R4b

Hardware for this step: K5 ×32

Make 8 complementary pairs of slats. Clamp a scrap jig with end planes 400 mm / 15.748 in apart and outside edges 240 mm / 9.449 in apart. End cuts are parallel, 26.66 degrees off square. Finished long-edge length 447.59 mm / 17.622 in; use 490 mm / 19.291 in rough blanks. Mark the intersection rhombus and remove 9 mm / 0.354 in from opposite faces. Test before batch cutting. End planes are 125 / 525 / 925 / 1325 / 1725 mm. One K5 screw per end, centred across each slat and 20 mm / 0.787 in vertically in from the cut plane. Clamp, pilot and drive flush. Glue only a well-fitted lap.

03 / Install the inward shelf cleats

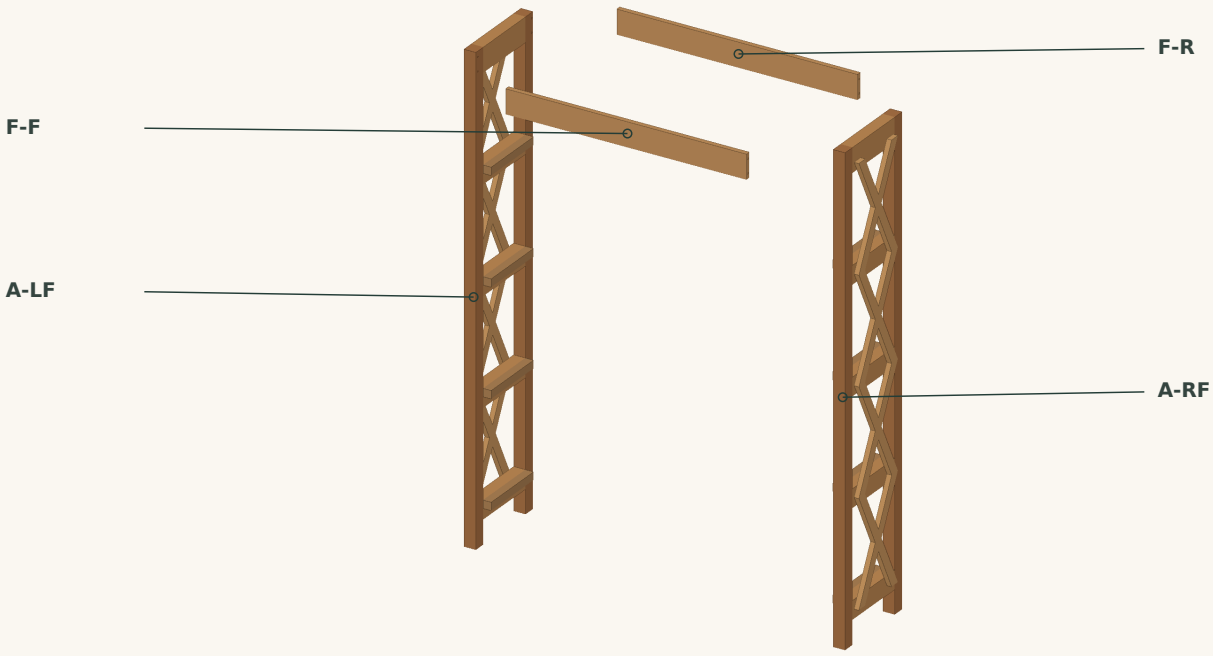


Part identifiers: A-LF, A-LR, B-L1, B-L2, B-L3, B-L4, C-L5, G-L1, G-L2, G-L3, G-L4, H-L1a, H-L1b, H-L2a, H-L2b, H-L3a, H-L3b, H-L4a, H-L4b, A-RF, A-RR, B-R1, B-R2, B-R3, B-R4, C-R5, G-R1, G-R2, G-R3, G-R4, H-R1a, H-R1b, H-R2a, H-R2b, H-R3a, H-R3b, H-R4a, H-R4b

Hardware for this step: K3 ×24

Keep G tops level with the B tops at 160 / 560 / 960 / 1360 mm. Cleats project 30 mm / 1.181 in into the opening. At each cleat drive K3 from its inside face toward B, at y = 110 / 175 / 240 mm and 15 mm / 0.591 in below its top. Do not move screws onto the X end screw lines. Check straightness and full bearing.

04 / Connect sides with the two aprons

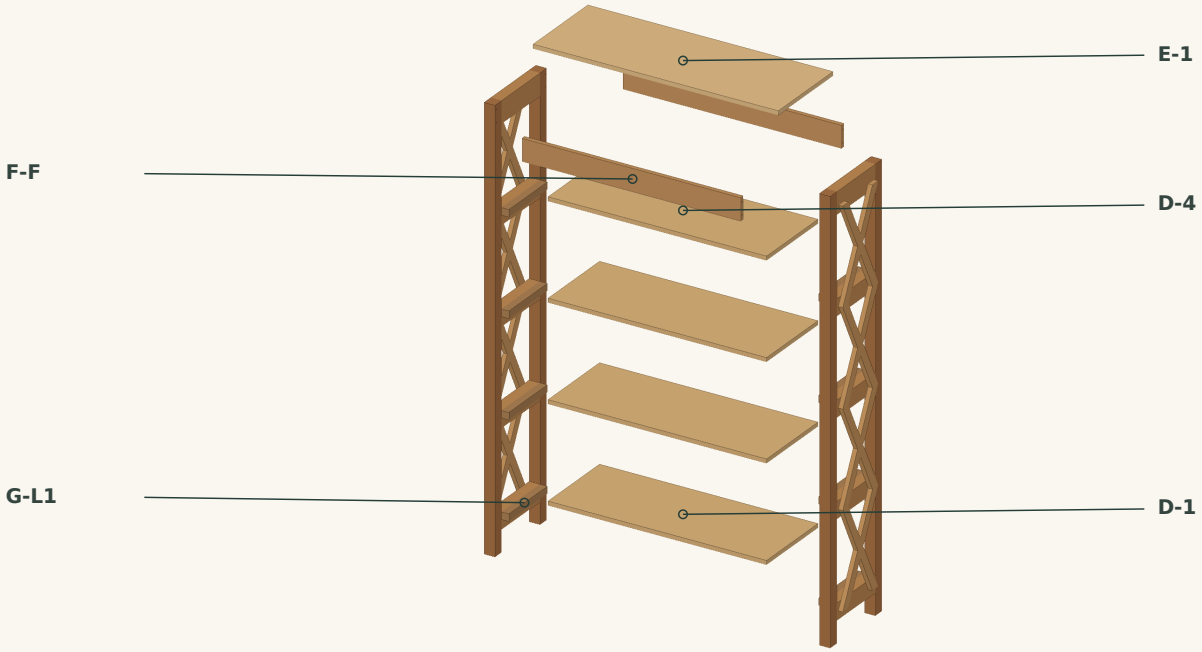


Part identifiers: A-LF, A-LR, B-L1, B-L2, B-L3, B-L4, C-L5, G-L1, G-L2, G-L3, G-L4, H-L1a, H-L1b, H-L2a, H-L2b, H-L3a, H-L3b, H-L4a, H-L4b, A-RF, A-RR, B-R1, B-R2, B-R3, B-R4, C-R5, G-R1, G-R2, G-R3, G-R4, H-R1a, H-R1b, H-R2a, H-R2b, H-R3a, H-R3b, H-R4a, H-R4b, F-F, F-R

Hardware for this step: K2 ×8

Use a helper. Maintain 910 mm / 35.827 in between inner post faces and 1000 mm / 39.370 in across the frame. Aprons occupy z = 1690–1782 mm; outer faces flush to front/rear posts. Two 6 mm / 0.236 in dowels per apron end: thickness centre 9 mm / 0.354 in, height offsets 20 / 72 mm. Drill 21 mm / 0.827 in deep for 20 mm / 0.787 in embed in each member. Dry-fit, glue and clamp square. Keep temporary support until cured and back bracing is installed.

05 / Secure the shelves and separate top

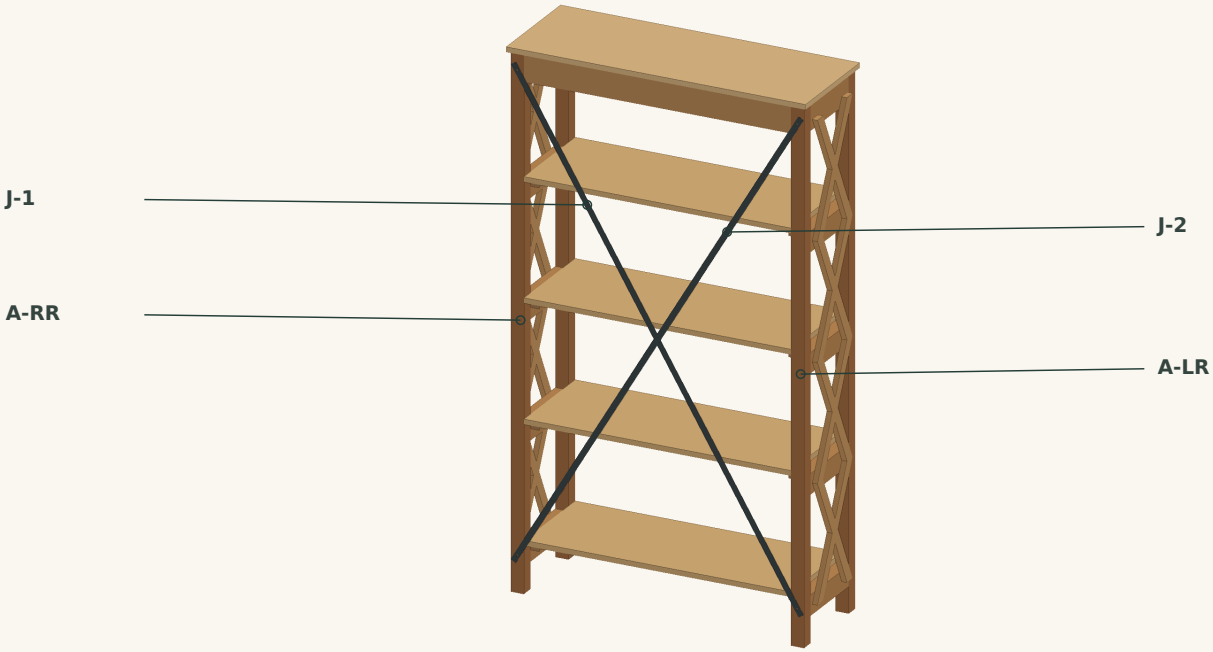


Part identifiers: A-LF, A-LR, B-L1, B-L2, B-L3, B-L4, C-L5, G-L1, G-L2, G-L3, G-L4, H-L1a, H-L1b, H-L2a, H-L2b, H-L3a, H-L3b, H-L4a, H-L4b, A-RF, A-RR, B-R1, B-R2, B-R3, B-R4, C-R5, G-R1, G-R2, G-R3, G-R4, H-R1a, H-R1b, H-R2a, H-R2b, H-R3a, H-R3b, H-R4a, H-R4b, D-1, D-2, D-3, D-4, E-1, F-F, F-R

Hardware for this step: K4 ×16, K8 ×4, K9 ×8

Shelf underside levels: 160 / 560 / 960 / 1360 mm; tops: 178 / 578 / 978 / 1378 mm. Maintain 0.5 mm / 0.020 in side clearance and 1 mm / 0.039 in front/rear setback. Four K4 per shelf: x = 60 / 940, y = 90 / 260 mm. Clearance-drill the shelf, pilot the cleat and countersink flush. Centre the top for 10 mm / 0.394 in overhang from the frame. Fix two K8 angles to the inner face of each apron at x = 272.5 / 727.5 mm. Use one K9 into the apron and one into the top per angle. Actual fitting is the drill template; verify tips cannot emerge.

06 / Brace the rear and anchor to the wall

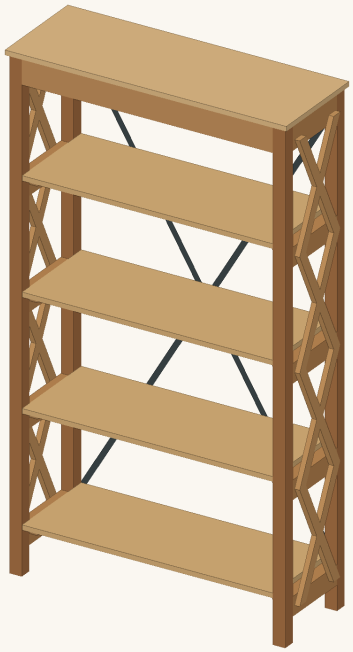


Part identifiers: A-LF, A-LR, B-L1, B-L2, B-L3, B-L4, C-L5, G-L1, G-L2, G-L3, G-L4, H-L1a, H-L1b, H-L2a, H-L2b, H-L3a, H-L3b, H-L4a, H-L4b, A-RF, A-RR, B-R1, B-R2, B-R3, B-R4, C-R5, G-R1, G-R2, G-R3, G-R4, H-R1a, H-R1b, H-R2a, H-R2b, H-R3a, H-R3b, H-R4a, H-R4b, D-1, D-2, D-3, D-4, E-1, F-F, F-R, J-1, J-2

Hardware for this step: K6 ×4, K7 ×2, K10 ×2

Square the case. Rear strap anchors are at post centres x = 22.5 / 977.5 mm and z = 125 / 1725 mm. Target hole pitch 1863.34 mm / 73.360 in; leave 25 mm / 0.984 in beyond each end hole. Drill clearance 5.5 mm / 0.217 in; deburr. Fit J-1 against the back, J-2 behind it with K7 washers at both ends. Use K6; snug without bowing the case. Fit two manufacturer-selected wall restraints near z = 1750 mm / 68.898 in. Avoid bores and concealed services; verify actual wall structure and screw paths. Do not load before restraint installation.

07 / Finish, level and inspect before use

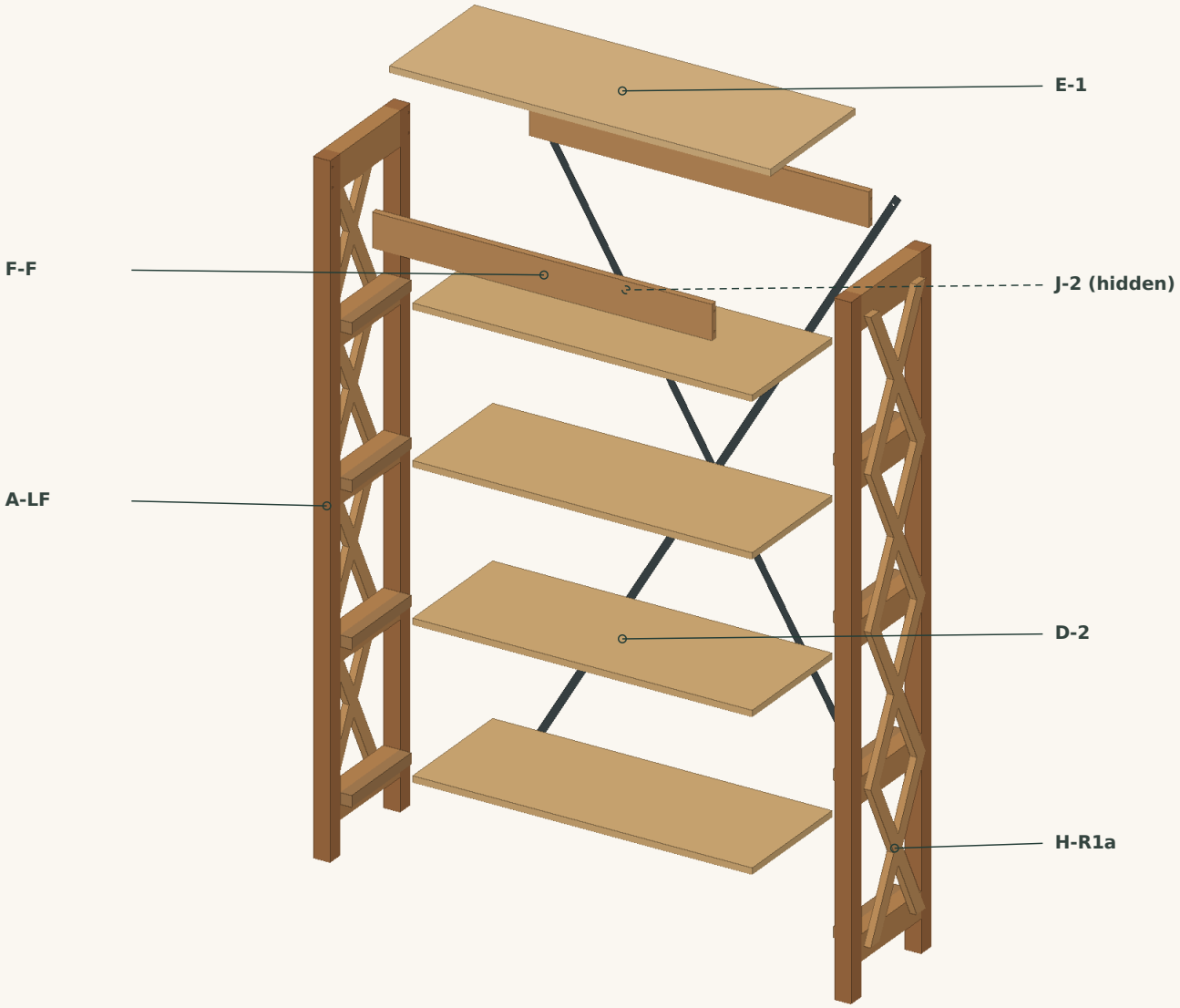


Part identifiers: A-LF, A-LR, B-L1, B-L2, B-L3, B-L4, C-L5, G-L1, G-L2, G-L3, G-L4, H-L1a, H-L1b, H-L2a, H-L2b, H-L3a, H-L3b, H-L4a, H-L4b, A-RF, A-RR, B-R1, B-R2, B-R3, B-R4, C-R5, G-R1, G-R2, G-R3, G-R4, H-R1a, H-R1b, H-R2a, H-R2b, H-R3a, H-R3b, H-R4a, H-R4b, D-1, D-2, D-3, D-4, E-1, F-F, F-R, J-1, J-2

Hardware for this step: K11 ×4

Ease accessible edges, sand to the finish maker’s guidance and keep glue faces clean. Cure all adhesive and finish fully. Fit equal non-slip pads and confirm four-foot bearing without rocking. Check every joint, shelf bearing, exposed screw and steel edge. Recheck restraints after relocation. Place heavier contents low; do not climb or sit on shelves. No safe working load has been established.

Complete exploded assembly



Part offsets are illustrative. Use the dimensions and sequence in this manual, not distances measured from this exploded drawing. The dashed J-2 callout identifies the strap behind the apron; see the rear assembly view for its visible connection. Connection/pilot holes and purchased brackets/restraints are not fully represented as CAD hardware.

Print, inspect & maintain

Print the manual on US Letter at a comfortable reading scale. The separate template_A4.pdf and template_Letter.pdf files are actual-size fabrication patterns: disable “fit”, print at 100%, check the 100 mm horizontal and 50 mm vertical bars, then trim and tape matching coordinates with the documented overlap.

Before use: check all four feet bear evenly; every shelf bears fully; no splits, loose dowels, open glue joints, proud screw tips or sharp steel remain. Confirm both restraints and sound wall structure according to the manufacturer. Recheck after relocation and whenever rocking or looseness develops.

Before commercial sale: commission an independent prototype build, verify selected materials and hardware, and obtain qualified strength, sustained-load creep, racking, impact, tip and installation testing. The plan assigns no safe working load. A dimension change requires renewed review; digital fit checks do not replace physical evidence.

Use guarded tools, clamps, dust extraction and suitable eye/hearing protection. Check for concealed services before wall drilling. Follow adhesive/finish ventilation and waste-handling guidance. Keep heavy contents low; do not climb or sit on the case.